School of
Information Technology
and Computer Science

CSCI101

Operating Systems

LAB 5 – WEEK 6

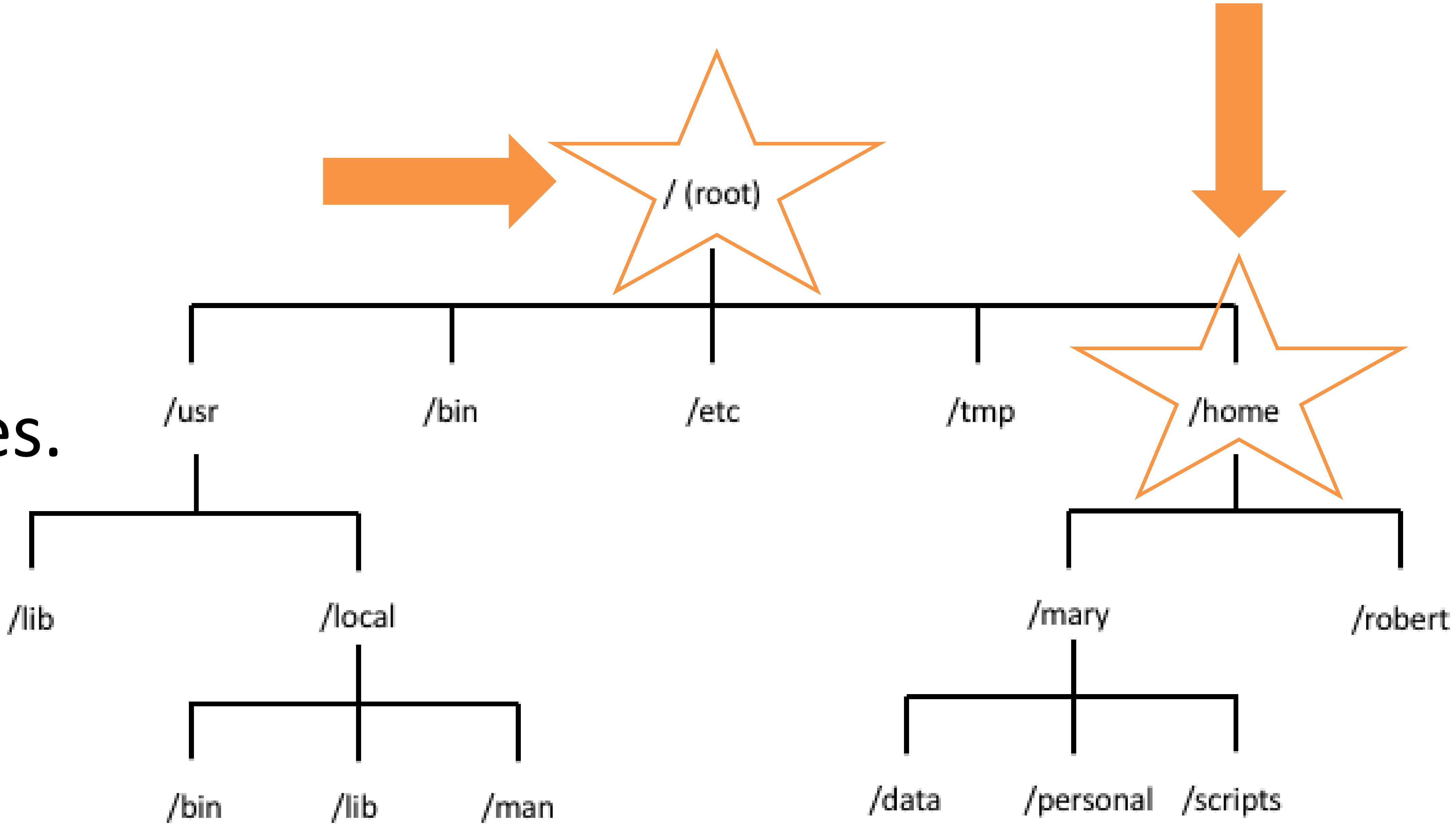
Nile University

CSCI101 Team

Recap on File System Structure

Key directories

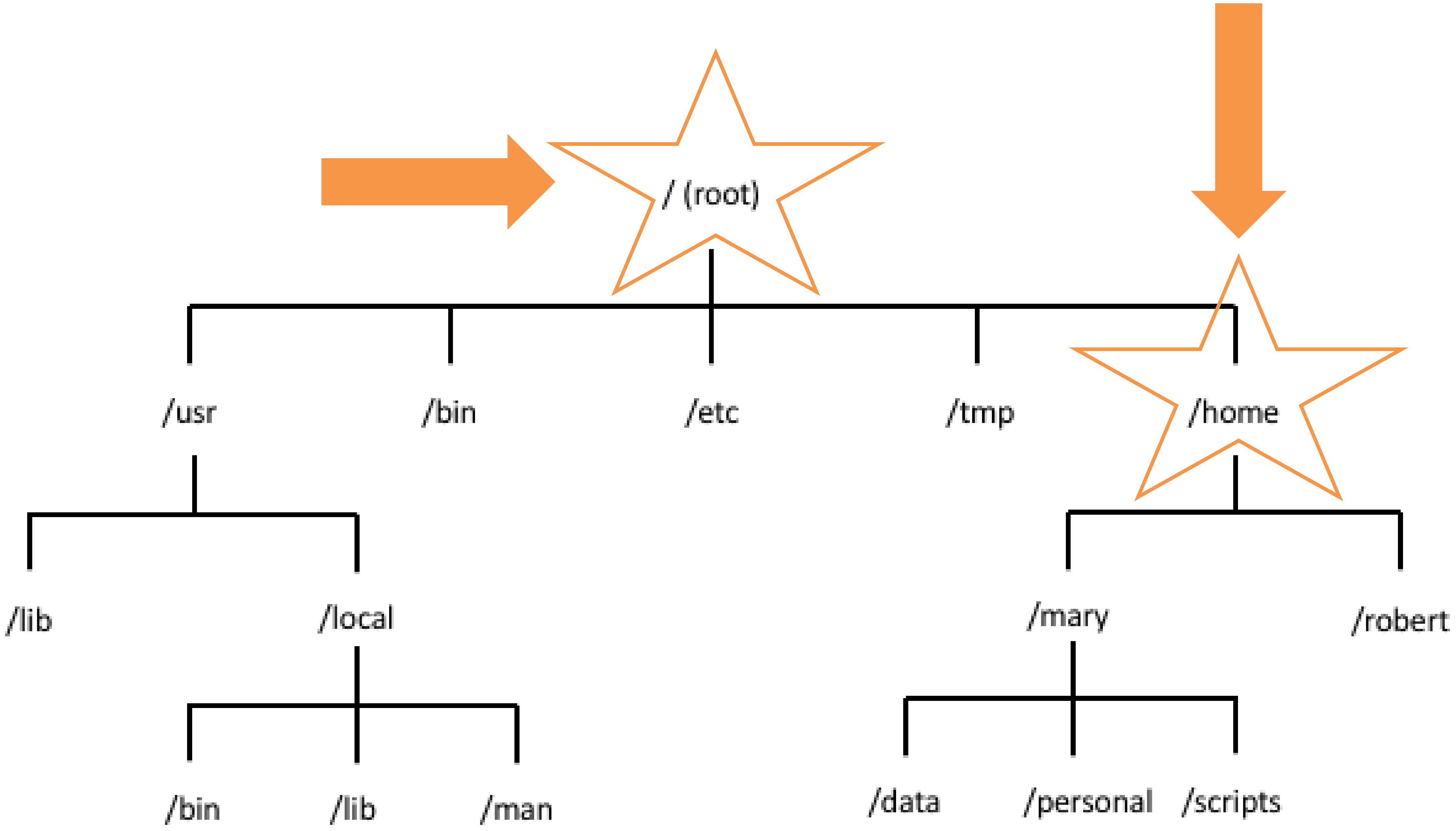
- / → Root directory.
- /home → User directories.



CSCI101

Lets Refresh! Question…

If I want to go to **mary** user directory what should I write in the path ?



The Basic Commands

Used for	Commands
Navigating Directories	ls → list files cd → change directory Pwd → print working directory
File Management	cp → copy mv → move rm → remove mkdir → make directory
Viewing Files	cat → read file less → read the first page in a file head → display first 5 rows in file tail → display last 5 rows in file
Creating File	touch
Getting Help about command	man --help

Question...

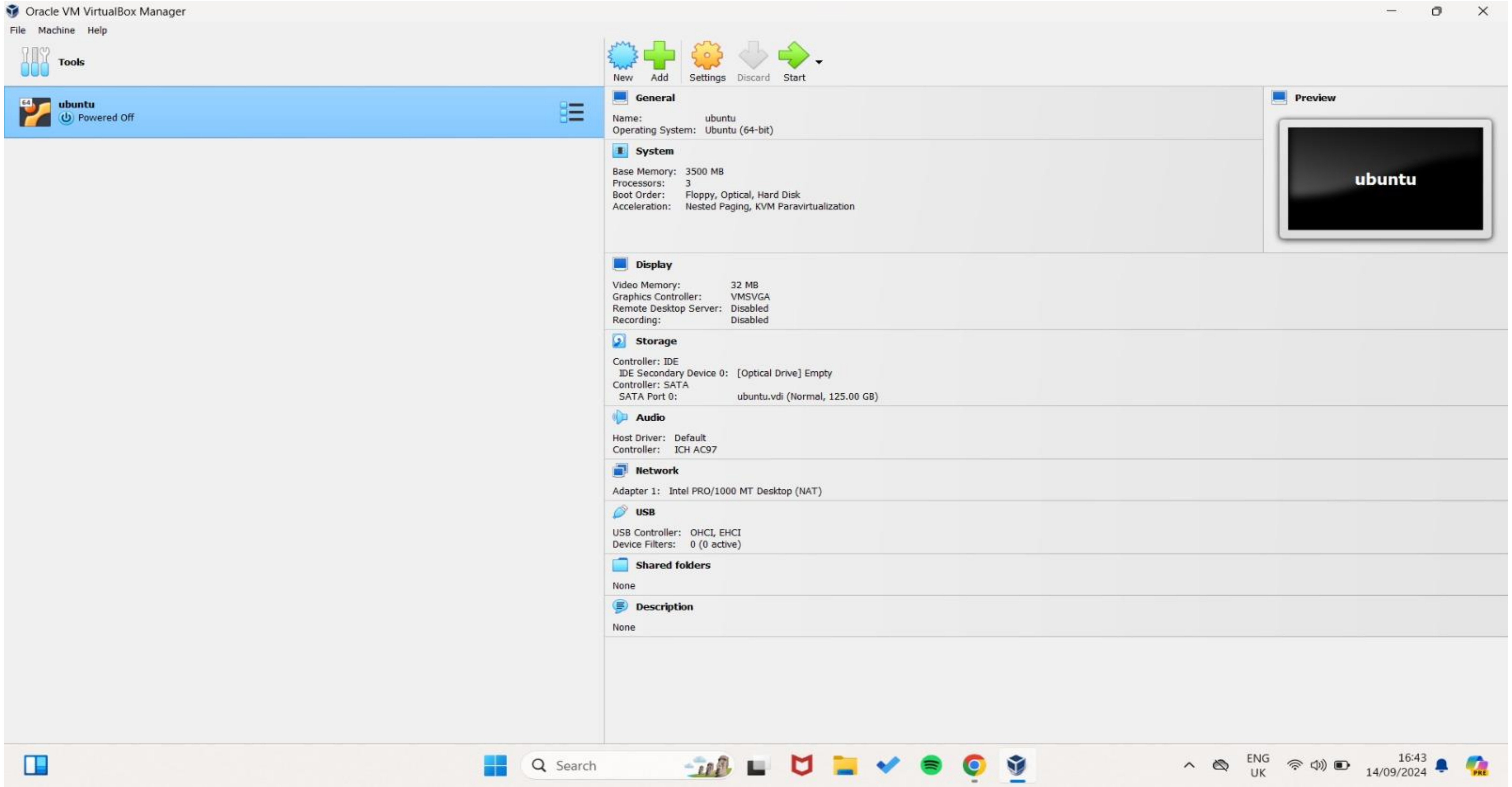
"Have you ever wondered how you could run two operating systems on the same computer at the same time without deleting anything?"

VirtualBox

Think of VirtualBox like a house that allows you to build many rooms, each with its own design. In our case, each ‘room’ is an operating system, and the ‘house’ is your actual computer

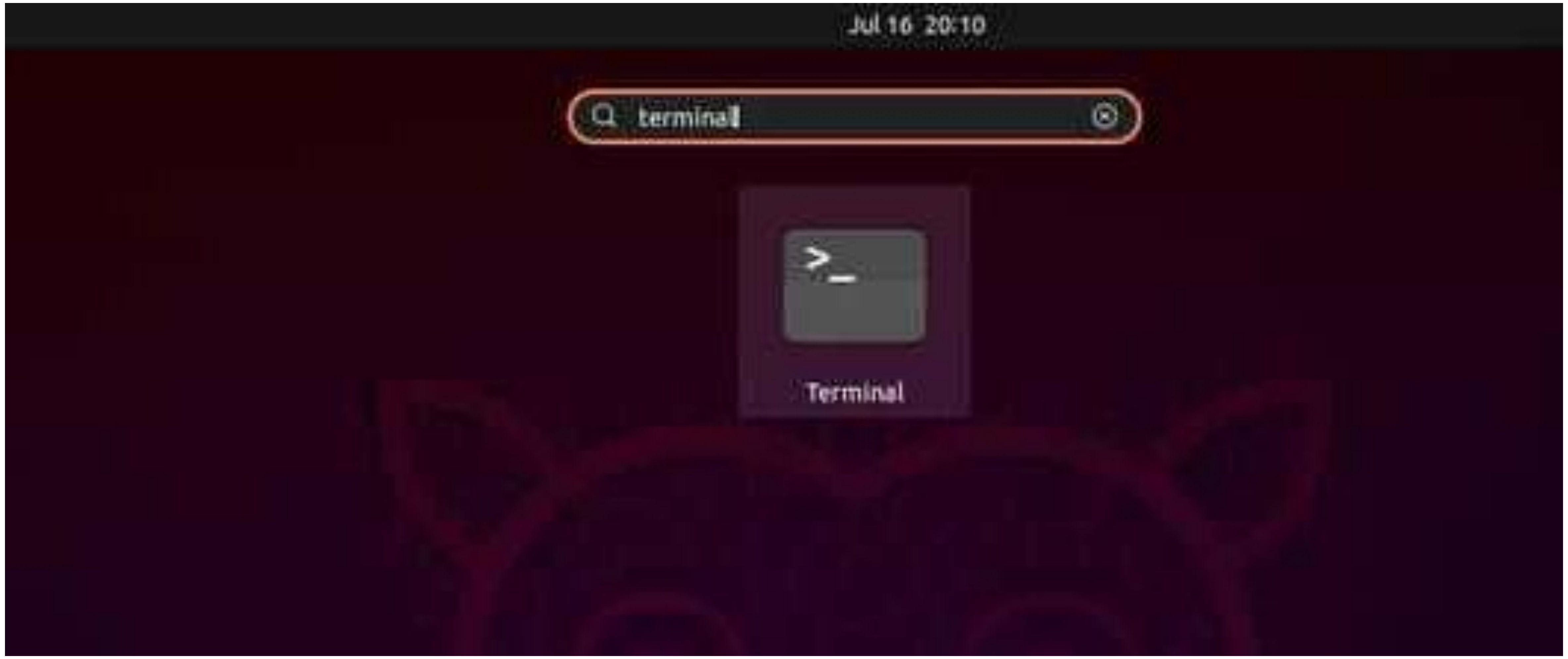
Definition -> VirtualBox is a software that allows you to run multiple operating systems, like Linux, on your existing computer, without the need to restart or delete anything.

VirtualBox Interface



Steps

Open your terminal in Linux



Steps

command: pwd – Print current working directory.

```
lenovo@lenovo-VMware-Virtual-Platform:~$ pwd
/home/lenovo
```

Steps

command: ls – List files in a directory.

```
lenovo@lenovo-VMware-Virtual-Platform:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
```

Steps

command: **cd** /path/to/directory – Change directories.
To go to Downloads for example:

```
lenovo@lenovo-VMware-Virtual-Platform:~$ cd Downloads
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$
```

How to return to home again?

1. Write → cd

```
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$ cd
lenovo@lenovo-VMware-Virtual-Platform:~$
```

Or

2. Write → cd ~

```
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$ cd ~
lenovo@lenovo-VMware-Virtual-Platform:~$
```

Steps

Can we move from root ‘/’ to Downloads directly? **No**

```
lenovo@lenovo-VMware-Virtual-Platform:~$ cd /
lenovo@lenovo-VMware-Virtual-Platform:/$ cd Downloads
bash: cd: Downloads: No such file or directory
```

we can by writing the whole path for the downloads like this:

```
lenovo@lenovo-VMware-Virtual-Platform:/$ cd /home/lenovo/Downloads
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$
```

Or just writing

cd ~/Downloads

Steps

command: mkdir – Create a directory.

```
lenovo@lenovo-VMware-Virtual-Platform:~$ mkdir course101
lenovo@lenovo-VMware-Virtual-Platform:~$ ls
course101  Documents  Music     Public  Templates
Desktop    Downloads  Pictures  snap    Videos
```

Look here folder created successfully

Steps

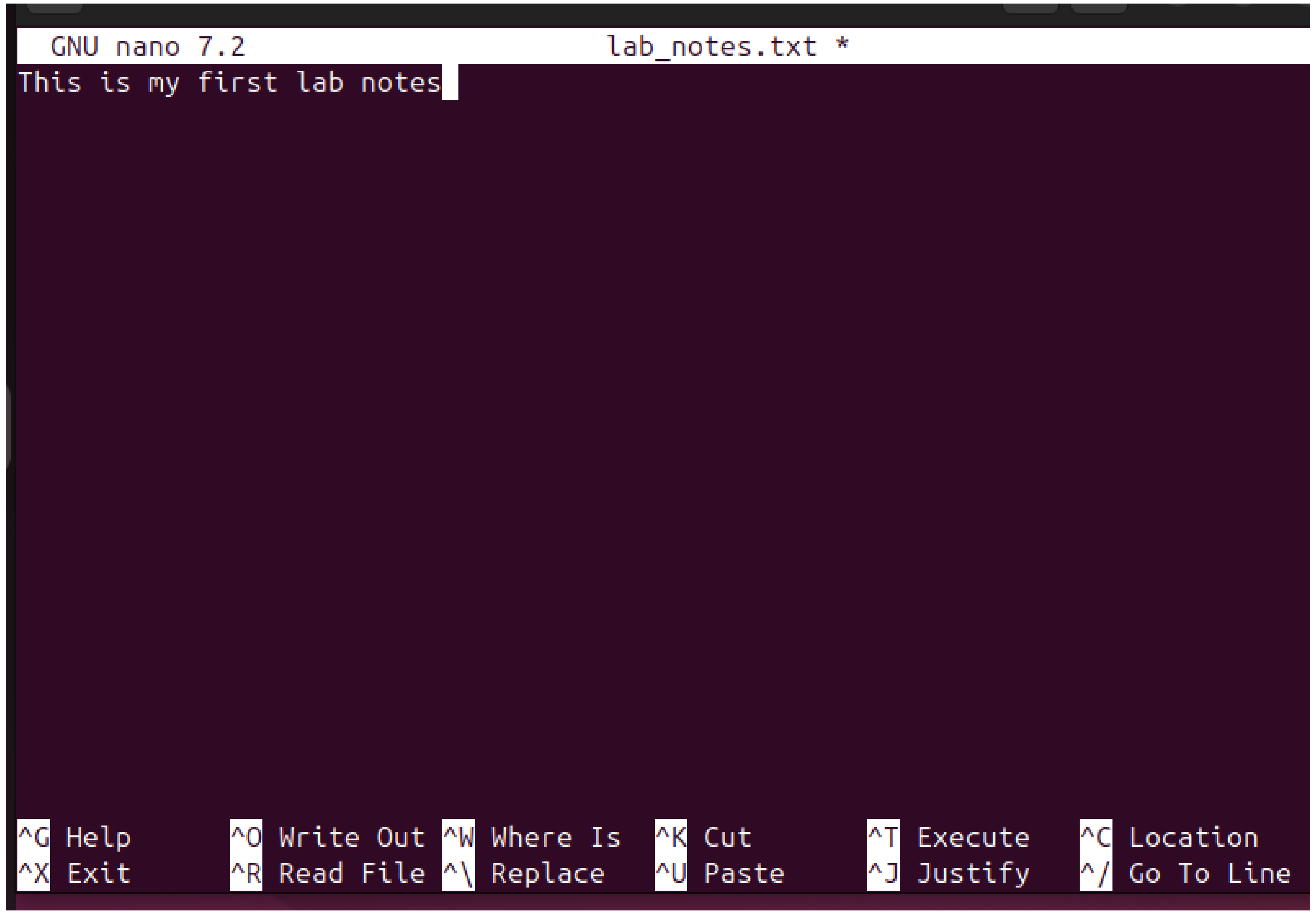
command: **touch** filename.txt – Create a file.

```
lenovo@lenovo-VMware-Virtual-Platform:~$ cd course101
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ touch lab_notes.txt
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ ls
lab_notes.txt
```

Look here file created successfully

How to Write in a File: Steps

1. **command: nano** filename.txt



Then enter

This page will appear

Write anything here

How to Write in a File: Steps

2. **command: echo** used to display text, but it can also be used to write to a file.

Ex1: echo "This is my second line in my notes" > lab_notes.txt

- This will create a file called lab_notes.txt and write **"This is my second line in my notes"** to it.
- If lab_notes.txt already exists, it will **overwrite** the content of the file.

```
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ echo "This is the second line in my notes" > lab_notes.txt
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ cat lab_notes.txt
This is the second line in my notes
```

Ex2: echo "This is my second line in my notes" >> lab_notes.txt

- To add content **without overwriting** the existing file, use >> instead of >.

```
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ echo "This is the second line in my notes" >> lab_notes.txt
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ cat lab_notes.txt
This is my first lab notes
This is the second line in my notes
```

How to View Content of a File: Steps

command: cat filename.txt , Also **head, tail** and **less** commands are used

```
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ cat lab_notes.txt
This is my first lab notes
```

Note commands differences

- cat - Display the contents of a file
- head - Display the **first** few lines of a file.
- tail - Display the **last** few lines of a file.
- less - View the contents of a file **page by page**.

How to View Content of a File: Steps

command: cp File.txt /copyLocation → Copy a file.

```
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ cp lab_notes.txt ~/Downloads
lenovo@lenovo-VMware-Virtual-Platform:~/course101$ cd ~/Downloads
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$ ls
lab_notes.txt
```

Look file copied from csci101 folder to Downloads successfully

command: mv File.txt /newLocation → Move a file.

```
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$ mv lab_notes.txt ~/Desktop
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$ ls
lenovo@lenovo-VMware-Virtual-Platform:~/Downloads$ cd ~/Desktop
lenovo@lenovo-VMware-Virtual-Platform:~/Desktop$ ls
lab_notes.txt
```

Look file moved from Downloads to Desktop successfully

command: rm File.txt → Delete a file.

```
lenovo@lenovo-VMware-Virtual-Platform:~/Desktop$ rm lab_notes.txt
lenovo@lenovo-VMware-Virtual-Platform:~/Desktop$ ls
lenovo@lenovo-VMware-Virtual-Platform:~/Desktop$
```

Look deleted successfully

How to Get Help For a Specific Command: Steps

using **man** → Display the manual for a command.

Syntax: man command_name

```
lenovo@lenovo-VMware-Virtual-Platform:~/Desktop$ man ls
```

```
LS(1)                              User Commands                              LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information about the FILEs (the current directory by default).  Sort entries alphabetically if
       none of -cftuvSUX nor --sort is specified.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

 Manual page ls(1) line 1 (press h for help or q to quit)
```

CSCI101

How to Get Help For a Specific Command: Steps

using **man** → Display the manual for a command.
Syntax: man command_name

```
lenovo@lenovo-VMware-Virtual-Platform:~/Desktop$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
```

Task 1

File Management task

1. Create an empty file called **Task.txt** inside your home directory.
2. Write **"Hello, World!"** into the file **Task.txt** using the echo command.
3. Append **"This is a test."** to the same file.
4. Copy **Task.txt** to Desktop
5. Remove **Task.txt** from your home directory. 👏🎆

Congratulations on finishing your task

Nile University
جـامعـة النيـل

School of
Information Technology
and Computer Science

CSCI101

Version Control (Git)

LAB 5 – WEEK 6

Nile University

CSCI101 Team

Creating Your GitHub Account

Go to github.com

Enter Your Email and Password

Welcome to GitHub!
Let’s begin the adventure

Enter your email*

✓ xixili6065@amxyy.com

Create a password*

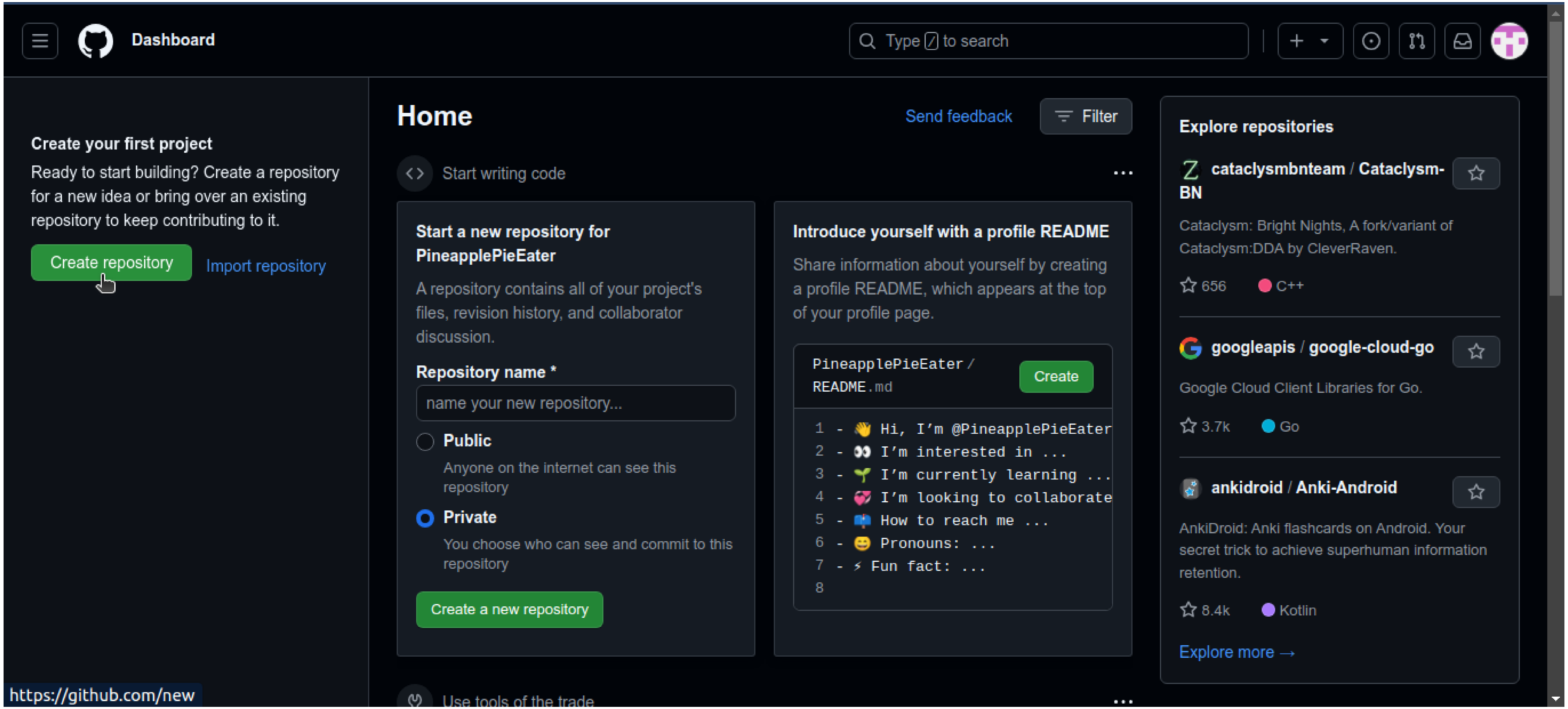
✓ P@ss123*3875/

Enter a username*

→ User422

Continue

Create a Repository

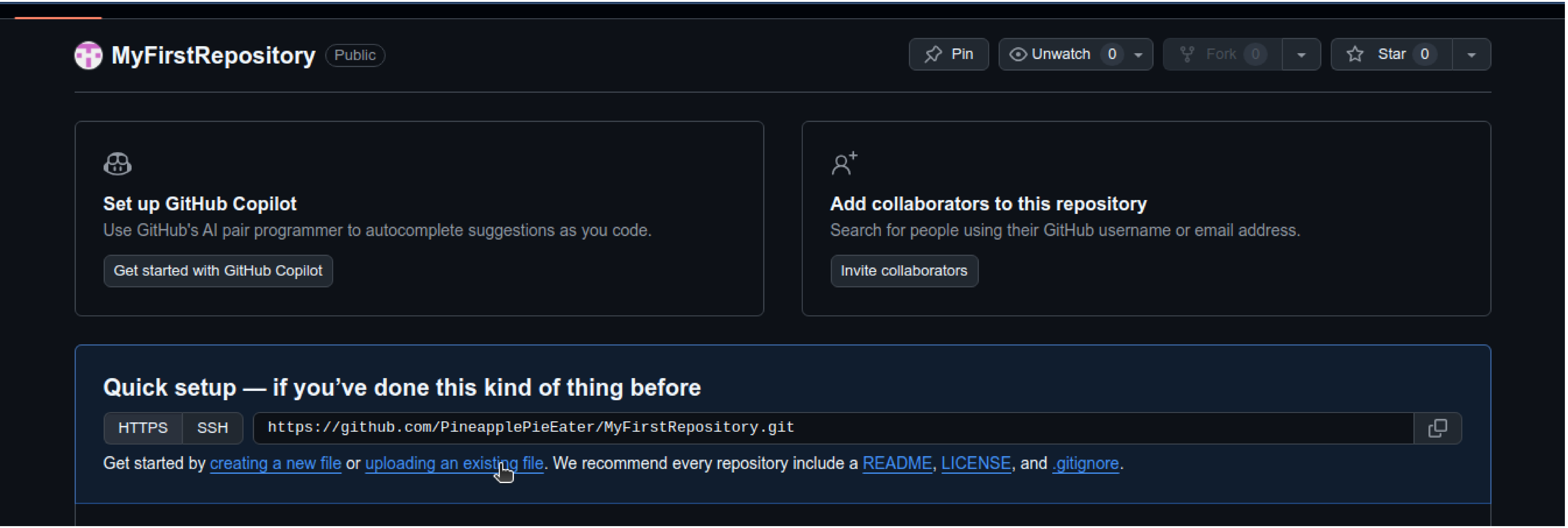


Create a Repository

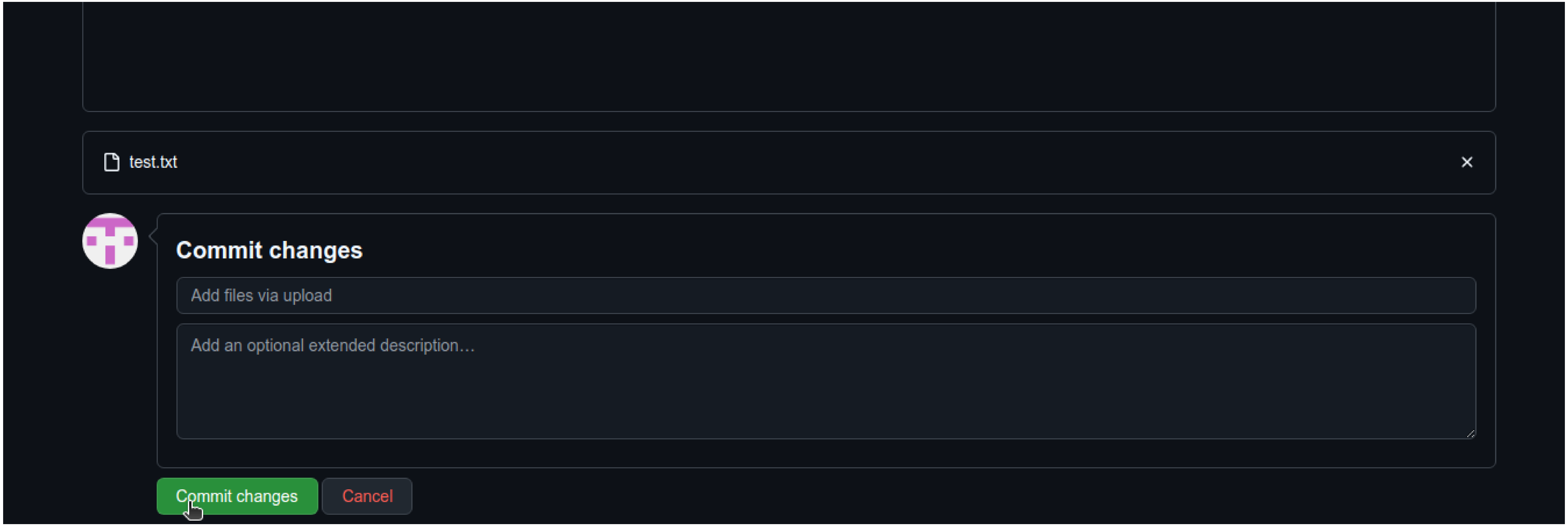
Name your repository & Create it

Upload Your Document to Your Repository

Choose your file and upload it



Commit Your Changes



Done!

CONGRATS! YOU MADE YOUR FIRST COMMIT

Task

- Make a pull request on “Task.txt” in this repository <repo_link>

Find the file with the name Task.txt

- Make a fork of this repository to your own github profile
- Find the Task.txt file on your newly created repository and edit it, adding your name and ID to the Task.txt file
- Finally, make a Pull request to add your edits to the main branch on this repository

THANK YOU

A n y Q u e s t i o n s ?